Haokai Ding

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Education

Shenzhen Technology University (SZTU) Sept 2022 –Jul 2026
B.Eng. in Electronic Science & Technology; GPA: **84.6/100**

Research focus: LIBS spectroscopy; robotics systems.

Tsinghua University —Shenzhen X-Institute Sept 2023 –Present
Tsien Excellence Engineering Program (TEEP), Joint Training (Underactuated Robotics)

Publications

An Underactuated Gripper with Grasshopper-Inspired Linkages and Delay Triggering for Pinching and Adaptive Grasping 2025
H. Ding†, Y. Chen†, D. Jia, W. Zhang*
ICIA 2025. [Accepted, EI]

A Novel Gripper with Semi-Peaucellier Linkage and Idle-Stroke Mechanism for Linear Pinching and Self-Adaptive Grasping 2025
H. Ding†, W. Zhang*
IROS 2025. [Accepted, EI]

Semi-Peaucellier Linkage and Differential Mechanism for Linear Pinching and Self-Adaptive Grasping 2025
H. Ding†, Z. Chen, T. Yang*, W. Zhang*
CASE 2025. [Published, EI]

Peaucellier Gripper: A Novel Underactuated Gripper for Linear Pinching and Self-Adaptive Grasp 2025
H. Ding†, J. Fan, Z. Zhu, Y. Zhao, K. Chen*, W. Zhang*
ARM 2025. [Published, EI]

Rapid Analysis of Heavy Metal Element Adsorption by SCG Based on LIBS Technology 2025
H. Xie, L. You, X. Fang, L. Li, **H. Ding**, Z. Zhou, G. Zhang, D. Zhang
E3S Web Conf., 615 (2025) 02010. [EI]

Recent Progress on the Research of 3D Printing in Aqueous Zinc-Ion Batteries 2025
Y. Liu†, **H. Ding**†, H. Chen, H. Gao, J. Yu, F. Mo, N. Wang*
Polymers 17(15):2136, 2025. [SCI, JCR Q1]

Legend: † Co-first author; * Corresponding author

Research & Projects

Underactuated Robotic Gripper Research —Shenzhen X-Institute, Tsinghua University 2023 –Present
Designed and tested underactuated robotic grippers with semi-Peaucellier linkage.

Focus: idle-stroke / delay-triggering mechanisms; validated prototypes via simulation and experiments.

LIBS Spectroscopy for Heavy Metal Analysis —SZTU 2022 –2023
Built and optimized LIBS system for rapid adsorption analysis.

Focus: improved calibration pipeline and enhanced acquisition accuracy.

Honors & Awards

ASME Student Mechanism & Robot Design Competition —Finalist 2025
International student competition finalist for mechanism and robot design.

President’ s Award, SZTU 2024
Highest student honor at Shenzhen Technology University.

China “Internet+” Innovation & Entrepreneurship (Guangdong) —Gold Prize 2023
Provincial-level first prize for innovation project.

“Challenge Cup” (Guangdong Division) —Second Prize 2023
Provincial student innovation competition award.

Chen Hsong Scholarship 2023
Dean’ s / Academic Excellence Scholarship, SZTU.

Skills

Programming: C/C++, Python, MATLAB
Tools: SolidWorks, LTSpice, Arduino, Linux
Languages: Chinese (native), English (fluent)